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Assignment 1 – Linear Algebra

Zia Afzal <zia@pucit.edu.pk> Fri, Jan 30, 2026 at 5:21 PM
To: bssef23ma@pucit.edu.pk

Dear Students,

This email is to inform you about **Assignment 1** for the course **Linear Algebra**. Please read the instructions carefully and ensure timely submission.

Question 1

Define the following terms clearly and concisely:

- System of *m* linear equations in *n* unknowns
- Matrix of coefficients
- Augmented matrix
- Consistent and inconsistent systems
- Elementary row operations
- Echelon form
- Reduced row echelon form
- General solution of a system
- Homogeneous system
- Trivial and non-trivial solutions of a homogeneous system
- Leading and free variables
- Pivot positions
- Pivot rows
- Pivot columns

Question 2

Explain the following methods in detail:

- Gaussian Elimination
- Gauss–Jordan Elimination

Question 3

Solve the following problems from **Exercise Set 1.2** of the prescribed textbook:

- Q2 (e)
- Q3 (b)
- Q5
- Q9
- Q19
- Q24 (d)
- Q27
- Q34
- Q42

Submission Deadline

The **deadline for submission** of Assignment 1 is:

Friday, 6 February 2026

Late submissions will not be entertained unless prior permission is granted.

If you have any questions or need clarification, feel free to contact me during office hours.

Best regards,
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